Reducing treatment time and number of nerves treated fails to improve cryoanalgesia outcomes

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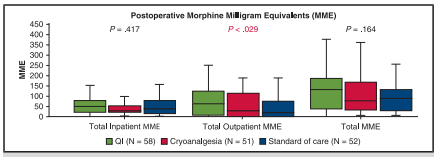
ABSTRACT

Objective: Cryoanalgesia is an attractive method to improve postoperative pain. However, our randomized trial did not show significant reduction in pain or opioid use in adults undergoing minimally invasive thoracic surgery. We hypothesized that reducing the number of intercostal spaces and treatment time may decrease postoperative pain and narcotic consumption.

Methods: In this quality improvement study, the quality improvement cohort received an intercostal nerve block with bupivacaine and lidocaine, and cryotherapy to 3 levels for 90 seconds. Outcomes from this cohort were compared with outcomes from our randomized trial cohorts: The cryoanalgesia cohort was treated with cryotherapy to 5 to 6 intercostal nerves for 120 seconds and intercostal nerve block, and the standard of care cohort was treated with intercostal nerve block only. Primary outcomes were opioid consumption up to the first postoperative visit measured by morphine milligram equivalents. Secondary outcomes included patient-reported pain and incentive spirometry volumes in the immediate postoperative period.

Results: A total of 161 patients were analyzed (quality improvement N = 58, standard of care N = 52, cryoanalgesia N = 51). There was no difference in baseline characteristics, procedure type, or length of stay. There were no differences in inpatient or total morphine milligram equivalents up to the first postoperative visit. Outpatient morphine milligram equivalents were significantly higher in the quality improvement cohort ($P = .029$). Patient-reported pain scores were similar among all groups, and there were no significant differences in incentive spirometry volumes.

Conclusions: Modifying the cryoanalgesia protocol to include fewer levels and less time did not improve opioid consumption, incentive spirometry, or pain scores in patients undergoing lung resection. (JTCVS Open 2026;29:101574)



Inpatient, outpatient, and total postoperative MME among cohorts.

CENTRAL MESSAGE

Applying cryoanalgesia to fewer intercostal levels with shorter treatment time does not reduce postoperative opioid consumption, patient-reported pain, or postoperative IS volumes.

PERSPECTIVE

In our randomized trial comparing cryoanalgesia with standard care in patients undergoing minimally invasive thoracic surgery, cryoanalgesia did not reduce pain or narcotic use. Patients in the cryoanalgesia group did experience significant neuropathy at 2 weeks. We conducted a QI study with a reduced cryoanalgesia protocol; there was no difference in pain or total opioid consumption.

Cryoanalgesia is an attractive strategy for managing postoperative pain. This technique induces Wallerian degeneration and allows the nerve to regain function over time. Several retrospective studies have reported significant improvement in acute postoperative pain in patients treated with cryoanalgesia.[1-5] We recently published the results of our randomized trial in patients undergoing minimally invasive thoracic surgery that did not show a reduction in pain or opioid use in the immediate postoperative period associated with cryoanalgesia.[6] In our trial, cryoanalgesia was administered to 5 to 6 intercostal nerves for 120 seconds each, as directed by AtriCure Inc.

Prior studies have demonstrated a dose-response relationship with cryoanalgesia, where longer duration of treatment

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Our Institutional Review Board did not require a specific informed consent for this study, but all patients whose data are included herein signed informed consent for cryoanalgesia as part of the Institutional Review Board study listed above.

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Abbreviations and Acronyms

CRYO = cryoanalgesia
INB = intercostal nerve block
IS = incentive spirometry
MME = morphine milligram equivalents
POD = postoperative day
QI = quality improvement
SOC = standard of care
VAS = Visual Analog Scale

is associated with prolonged numbness and potential neuropathy.[7,8] Prolonged treatment time also has the potential to cause nerve ablation and permanent damage.[9] Cryoanalgesia protocols vary among studies, and some studies show that a significant reduction in acute postoperative pain has been associated with cryoanalgesia administered for 60 and 120 seconds.[10-12] Conversely, other groups have reported no improvement in acute pain after treatment durations of 60 or 90 seconds.[3,13,14]

Given the results from our negative randomized trial and lack of a uniform protocol for administering cryoanalgesia based on the present literature, we conducted a quality improvement (QI) study to evaluate whether decreasing the treatment time and number of nerves treated with cryoanalgesia would reduce acute postoperative pain and opioid use. We treated patients undergoing minimally invasive thoracic surgery with cryoanalgesia to 3 intercostal nerves for 90 seconds each and compared patient-reported pain and morphine milligram equivalents (MME) with the cryoanalgesia (CRYO) cohort and standard of care (SOC) cohort from our randomized trial.

PATIENTS AND METHODS

This was a QI study following the outcomes of our randomized trial.[6] Inclusion and exclusion criteria were consistent with those in our trial (NCT05348447, Study No. 021-238, approved on November 9, 2021).[6] In short, patients at least 18 years of age undergoing elective minimally invasive thoracic surgery were included. The QI cohort included treatment of 73 consecutive patients undergoing elective robotic pulmonary resections. Fifteen patients were excluded due to opioid or gabapentin use. Patients randomized to the SOC or CRYO cohort underwent video-assisted thoracoscopic surgery or robotic surgery. Patients with a history of chronic narcotic or gabapentin use, substance abuse, fibromyalgia, Child-Pugh B or C cirrhosis, and patients on dialysis were excluded. The QI cohort was treated with an intercostal nerve block (INB) using lidocaine and bupivacaine to nerves 2 to 10 and cryoanalgesia to 3 intercostal levels for 90 seconds each at −80 °C. Cryoanalgesia was administered using the Cryos-L CryoSphere probe and CryoICE system (AtriCure Inc). Patients randomized to the SOC cohort were treated with the same INB. Patients randomized to the CRYO cohort were treated with INB and cryoanalgesia to 5 to 6 intercostal nerves for 120 seconds each at −80 °C. Study design, surgical procedure, and pain management were previously reported.[6]

Our Institutional Review Board did not require a specific informed consent for the QI study, but all patients signed informed consent for cryoanalgesia in the preoperative office visit.

OUTCOMES

The primary outcome of interest was postoperative opioid use. This was measured as MME. Conversion factors were 6.7 for hydromorphone, 3 for intravenous morphine, 0.1 for fentanyl, 0.1 for oral tramadol, and 1.5 for oxycodone to calculate total MME. Total inpatient MME included narcotics administered in the postanesthesia care unit up until the time of discharge. Total outpatient MME was measured from discharge up until the first postoperative visit. Patients brought medication bottles to the first postoperative visit, and remaining pills were tallied by the research staff. Total MME was the sum of total inpatient and outpatient MME.

Secondary outcomes included patient-reported pain on postoperative days (PODs) 1 and 2, and improvement in incentive spirometry (IS) from baseline in the immediate postoperative period. Pain was assessed by the Visual Analog Scale (VAS), with 0 indicating no pain and 10 indicating maximum pain.[15] Pain was assessed at multiple timepoints throughout the first and second POD by nursing staff. The average pain score over 24 hours was used for analysis. Baseline IS volumes were measured at the preoperative clinic visit. The best of 3 attempts at the preoperative visit and on PODs 1 and 2 were used for analysis.

Statistical Analysis

Continuous variables were reported as median and interquartile range (25th-75th percentile). Categorical variables were reported as frequencies and percentages. Kruskal–Wallis test was used to analyze the comparison between continuous variables. Pearson's chi-square test was used to analyze the comparison between categorical variables. Data were analyzed using IBM SPSS Statistics for MAC Version 29.0 (SPSS Inc).

RESULTS

A total of 161 patients were analyzed, of whom 58 patients comprised the QI cohort, and 52 and 51 patients were randomized to SOC and CRYO cohorts, respectively. There was no difference in age, gender, minimally invasive technique, hospital length of stay, or time from discharge to the first postoperative visit between the groups (Table 1). When comparing the difference between preoperative IS volumes and POD 1 IS volumes, all 3 groups had a median difference of 750 mL. The difference between baseline and POD 2 IS

TABLE 1. Patient and postoperative characteristics in the immediate postoperative period

Variable	QI cohort (N = 58)	CRYO cohort (N = 51)	SOC cohort (N = 52)	*P*
Age (y)	69 (63.3-73.0)	66 (61.0-71.0)	70.5 (63.3-75.8)	.105
Female	32 (55.2)	29 (56.9)	32 (61.5)	.787
Anatomic lung resection	50 (86.2)	45 (88.2)	46 (88.5)	.924
Hospital length of stay (d)	2.0 (1.0-4.0)	2.0 (1.0-3.0)	2.5 (1.0-4.8)	.697
Δ_{IS} POD 1 (mL)	750 (500-1250)	750 (500-1250)	750 (500-1000)	.437
Δ_{IS} POD 2 (mL)	1000 (375-1625)	750 (-187-1437)	1000 (250-1150)	.366
VAS pain score POD 1	3.5 (2.3-5.0)	3.8 (2.0-6.0)	4.1 (2.1-6.0)	.094
VAS pain score POD 2	3.1 (2.1-4.8)	3.5 (1.0-5.3)	2.0 (1.0-5.3)	.834

Values are reported as frequency (%) or median (interquartile range). *QI,* Quality improvement; *CRYO,* cryoanalgesia; *SOC,* standard of care; Δ_{IS}, difference between baseline and postoperative IS volume; *POD,* postoperative day; *VAS,* Visual Analog Scale.

volumes was 1000 mL for both the QI and SOC cohorts, and there was no difference between any of the groups. There was no difference in pain scores on POD 1. Median VAS scores were 3.5, 3.8, and 4.1 for the QI, CRYO, and SOC cohorts, respectively ($P = .094$) (Table 1). Median VAS pain scores on POD 2 were lower in all groups, and there was no statistically significant difference ($P = .834$) (Table 1).

Postoperative MME was compared between groups in the inpatient setting, at outpatient follow-up, and as a combined total of inpatient and outpatient use. Inpatient MME was not statistically different among the groups with a median of 58.3 for QI, compared with 44.9 and 38.4 for the SOC and CRYO groups, respectively ($P = .417$) (Figure 1). Outpatient MME was significantly higher for patients in the QI cohort, with a median MME of 75, 22.5, and 32.5 for patients in the QI, SOC, and CRYO cohorts, respectively ($P = .029$) (Figure 1). Median total MME was 156.5 for QI, 105.9 for SOC, and 94.6 for CRYO ($P = .164$) (Figure 1).

DISCUSSION

In this QI study, we reduced the number of intercostal nerves treated with cryotherapy as well as treatment time to evaluate whether pain and opioid use were significantly reduced in patients undergoing minimally invasive thoracic surgery. The rationale for this study was based on our randomized trial that did not show a reduction in pain or opioid use in the acute postoperative period but did show a significant increase in neuropathic pain in patients treated with cryoanalgesia. After application of our QI cryotherapy protocol, there was no significant difference in patient-reported VAS pain scores up through POD 2 and total MME measured from the time of surgery to the first postoperative visit was not significantly reduced. However, when specifically comparing outpatient MME, patients in the QI cohort consumed significantly more narcotics from the time of discharge to the first postoperative follow-up visit.

Cryoanalgesia has been most commonly studied in pediatric patients undergoing pectus excavatum repair and in the adult population undergoing thoracotomy. Randomized trials and retrospective studies have shown reduced pain on PODs 1 and 2 associated with cryoanalgesia after pectus repair.[2,4,16,17] Similar results have been observed in adult patients after thoracotomy.[7,11] In contrast, randomized trials by our group and others have not shown a significant reduction in acute postoperative pain associated with cryoanalgesia.[6,10] The present study confirms these findings because there was no significant improvement in pain in the QI cohort compared with the SOC and CRYO cohorts. This variability in outcomes across studies may be related to the analgesic strategy in the comparative group and surgical approach.

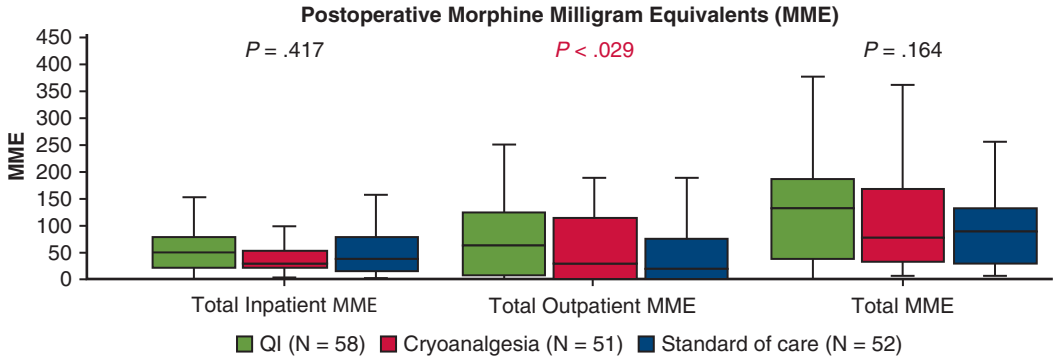

FIGURE 1. Comparison of inpatient, outpatient, and total postoperative MME across cohorts.

Epidural analgesia historically has been the technique of choice for managing post-thoracotomy pain.[18] In studies that compared cryoanalgesia with epidural in patients undergoing thoracotomy, there has been no difference in acute pain associated with cryoanalgesia.[13,14] In contrast, studies that report an improvement in acute pain associated with cryoanalgesia have compared outcomes with patients treated with parenteral analgesia.[7,11] In regard to surgical technique, minimally invasive thoracic surgery is associated with reduced postoperative pain compared with thoracotomy.[19] In contrast to thoracotomy, recommended analgesia for minimally invasive thoracic surgery includes regional analgesia to the thoracic nerves.[20] It is possible that in our randomized trial and this QI study, cryoanalgesia did not result in superior outcomes because our comparative group received INBs. Our group was the first to randomize patients undergoing minimally invasive thoracic surgery to receive cryoanalgesia and compare those outcomes with a modified cryotherapy protocol. As such, published data in this patient cohort and a comparison of our results to other studies are limited. Additional studies that compare cryoanalgesia with varying methods of analgesia in the minimally invasive setting are needed.

Several studies have reported reduced MME after cryoanalgesia. In a retrospective study by O'Connor and colleagues,[12] discharge prescription MME was significantly reduced in patients treated with cryoanalgesia who underwent minimally invasive thoracic surgery and open thoracotomy. Although there was no difference in inpatient opioid use, opioid refills for patients treated with cryoanalgesia were lower by a mean of 88 mg MME compared with patients in the control group who were treated with narcotics and nonnarcotics.[12] Sepsas and colleagues[11] reported a significant reduction in morphine use for the first 96 hours postoperatively as well as reduced narcotics up through 2 weeks postoperatively. In pediatric patients, several groups have reported cryoanalgesia significantly lowered inpatient opioid use and MMEs prescribed at discharge.[10,13,16,17,21,22] In contrast, Pilkington and colleagues[4] did not report a difference in inpatient opioid use or narcotic discharge prescriptions. In our study, we did not observe a reduction in narcotic use associated with cryoanalgesia. We did observe an increase in outpatient MME in patients treated with cryoanalgesia. It is possible that differences in patient populations, cryoanalgesia technique, and surgical technique may affect MME outcomes across studies.

Cryoanalgesia treatment protocols vary by institution in terms of cryo-probe, cooling temperature, treatment duration, and number of nerves ablated. However, a preclinical model by Guitron[23] showed that Wallerian degeneration occurs in the majority of nerves treated regardless of treatment time that ranged from 30 to 180 seconds. In a clinical study, Zacha and colleagues[9] treated intercostal nerves for 60 or 120 seconds and reported there was no difference in pain up through POD 14 between both treatment times. With regard to temperature, axonotmesis occurs between temperatures of −20 °C and −100 °C, whereas temperatures below −100 °C have the potential to cause irreversible nerve damage.[23] Published studies have applied cryotherapy for 30 to 120 seconds at temperatures of −20 °C to −80 °C with varying outcomes. In a study by Ju and colleagues,[14] nerves were treated at −70 °C for 90 seconds each. The authors report no difference in patient-reported pain in the first 3 PODs after thoracotomy. Moorjani and colleagues[7] treated intercostal nerves for 30 to 120 seconds at −50 °C. In this study, patients treated with cryoanalgesia reported significantly less pain over the first 7 PODs and used significantly less opiates over this time period.[7] In the retrospective study by O'Connor and colleagues,[12] patients were treated between −50 °C and −70 °C for 120 seconds, which resulted in a significant reduction in pain and opioid use in the acute postoperative period.[12] Similar results were reported by Sepsas and colleagues,[11] who observed a reduction in pain and opioid use after treatment with cryoanalgesia at −40 °C for 60 seconds. In contrast, Gwak and colleagues[24] reported no difference in pain or opioid use in patients treated with cryoanalgesia at −20 °C for 90 seconds up through 7 days after thoracotomy. In a pediatric randomized trial by Graves and colleagues,[10] patients treated with cryoanalgesia at −60 °C for 120 seconds used significantly fewer narcotics up through POD 3, but there was no statistical difference in patient-reported pain. Overall, these data suggest additional factors beyond cryotherapy protocol influence outcomes, and further studies with standardized cryoanalgesia protocols are needed to identify variables contributing to reduced pain and MME associated with cryoanalgesia.

Limitations

This study has limitations. The study was not powered to detect a difference in opioid use among the 3 groups; however, the SOC and CRYO cohorts were powered from the original trial to detect a reduction in total MME associated with cryoanalgesia. Patients in the QI cohort were not blinded to their intervention, but results from our randomized trial were not yet analyzed and therefore not able to be shared with patients in the QI cohort. Reducing treatment time to 90 seconds for 3 intercostal nerves was chosen based on results from our randomized trial and treatment protocols from current literature. Therefore, the treatment protocol for the QI cohort was not based solely on scientific rationale.

CONCLUSIONS

Applying cryoanalgesia to fewer nerves for less time does not significantly reduce patient-reported pain in the acute postoperative period. Inpatient, outpatient, and total MME were highest in the QI cohort. MME from the time of surgery up through the first postoperative visit was significantly higher in the QI cohort. There was no significant improvement in hospital length of stay or IS volumes on

PODs 1 and 2 compared with baseline. Based on these outcomes, reducing cryotherapy treatment time to 90 seconds for 3 intercostal nerves is not recommended in patients undergoing minimally invasive thoracic surgery.

Conflict of Interest Statement

B.W. is a proctor for Intuitive Surgery and a speaker for AstraZeneca and Merck. All other authors reported no conflicts of interest.

The *Journal* policy requires editors and reviewers to disclose conflicts of interest and to decline handling or reviewing manuscripts for which they may have a conflict of interest. The editors and reviewers of this article have no conflicts of interest.

References

1. Tanaka A, Al-Rstum Z, Leonard SD, et al. Intraoperative intercostal nerve cryoanalgesia improves pain control after descending and thoracoabdominal aortic aneurysm repairs. *Ann Thorac Surg*. 2020;109(1):249-254. https://doi.org/10.1016/j.athoracsur.2019.07.083
2. Rim GM, Kim HK, Koo JM, Park HJ. A randomized controlled trial of cryoanalgesia for pain management following pectus excavatum repair: a single-center, single-blind, parallel design study. *Eur J Pediatr Surg*. 2024;34(4):338-345. https://doi.org/10.1055/a-2117-4628
3. Yang MK, Cho CH, Kim YC. The effects of cryoanalgesia combined with thoracic epidural analgesia in patients undergoing thoracotomy. *Anaesthesia*. 2004;59(11):1073-1077. https://doi.org/10.1111/j.1365-2044.2004.03896.x
4. Pilkington M, Harbaugh CM, Hirschl RB, Geiger JD, Gadepalli SK. Use of cryoanalgesia for pain management for the modified Ravitch procedure in children. *J Pediatr Surg*. 2020;55(7):1381-1384. https://doi.org/10.1016/j.jpedsurg.2019.09.016
5. Lau WC, Shannon FL, Bolling SF, et al. Intercostal cryo nerve block in minimally invasive cardiac surgery: the prospective randomized FROST trial. *Pain Ther*. 2021;10(2):1579-1592. https://doi.org/10.1007/s40122-021-00318-0
6. Weksler B, Maxwell C, Drake L, et al. A randomized study of cryoablation of intercostal nerves in patients undergoing minimally invasive thoracic surgery. *J Thorac Cardiovasc Surg*. 2024;69(5):1375-1382.e1. https://doi.org/10.1016/j.jtcvs.2024.10.058
7. Moorjani N, Zhao F, Tian Y, Liang C, Kaluba J, Maiwand MO. Effects of cryoanalgesia on post-thoracotomy pain and on the structure of intercostal nerves: a human prospective randomized trial and a histological study. *Eur J Cardiothorac Surg*. 2001;20(3):502-507. https://doi.org/10.1016/s1010-7940(01)00815-6
8. Beazley RM, Bagley DH, Ketcham AS. The effect of cryosurgery on peripheral nerves. *J Surg Res*. 1974;16(3):231-234. https://doi.org/10.1016/0022-4804(74)90036-5
9. Zacha S, Lopez M, Bilas J, Skonieczna-Żydecka K, Miegoń J, Biernawska J. One- versus two-minute intercostal nerve cryoanalgesia in children undergoing surgery for funnel chest deformity. *J Pers Med*. 2024;14(8):875. https://doi.org/10.3390/jpm14080875
10. Graves CE, Moyer J, Zobel MJ, et al. Intraoperative intercostal nerve cryoablation during the Nuss procedure reduces length of stay and opioid requirement: a randomized clinical trial. *J Pediatr Surg*. 2019;54(11):2250-2256. https://doi.org/10.1016/j.jpedsurg.2019.02.057
11. Sepsas E, Misthos P, Anagnostopulu M, Toparlaki O, Voyagis G, Kakaris S. The role of intercostal cryoanalgesia in post-thoracotomy analgesia. *Interact Cardiovasc Thorac Surg*. 2013;16(6):814-818. https://doi.org/10.1093/icvts/ivs516
12. O'Connor LA, Dua A, Orhurhu V, Hoepp LM, Quinn CC. Opioid requirements after intercostal cryoanalgesia in thoracic surgery. *J Surg Res*. 2022;274:232-241. https://doi.org/10.1016/j.jss.2022.01.009
13. Mustola ST, Lempinen J, Saimanen E, Vilkko P. Efficacy of thoracic epidural analgesia with or without intercostal nerve cryoanalgesia for postthoracotomy pain. *Ann Thorac Surg*. 2011;91(3):869-873. https://doi.org/10.1016/j.athoracsur.2010.11.014
14. Ju H, Feng Y, Yang BX, Wang J. Comparison of epidural analgesia and intercostal nerve cryoanalgesia for post-thoracotomy pain control. *Eur J Pain*. 2008;12(3):378-384. https://doi.org/10.1016/j.ejpain.2007.07.011
15. Bijur PE, Silver W, Gallagher EJ. Reliability of the visual analog scale for measurement of acute pain. *Acad Emerg Med*. 2001;8(12):1153-1157. https://doi.org/10.1111/j.1553-2712.2001.tb01132.x
16. Aiken TJ, Stahl CC, Lemaster D, et al. Intercostal nerve cryoablation is associated with lower hospital cost during minimally invasive Nuss procedure for pectus excavatum. *J Pediatr Surg*. 2021;56(10):1841-1845. https://doi.org/10.1016/j.jpedsurg.2020.10.009
17. Song SH, Moon DH, Shim YH, Jung H, Lee S. Limited cryoablation reduces hospital stay and opioid consumption compared to thoracic epidural analgesia after minimally invasive repair of pectus excavatum. *Medicine (Baltimore)*. 2022;101(31):e29773. https://doi.org/10.1097/MD.0000000000029773
18. Lubenow TR, Faber LP, McCarthy RJ, Hopkins EM, Warren WH, Ivankovich AD. Postthoracotomy pain management using continuous epidural analgesia in 1,324 patients. *Ann Thorac Surg*. 1994;58(4):924-929; discussion 929-930, https://doi.org/10.1016/0003-4975(94)90435-9
19. Bendixen M, Jørgensen OD, Kronborg C, Andersen C, Licht PB. Postoperative pain and quality of life after lobectomy via video-assisted thoracoscopic surgery or anterolateral thoracotomy for early stage lung cancer: a randomised controlled trial. *Lancet Oncol*. 2016;17(6):836-844. https://doi.org/10.1016/S1470-2045(16)00173-X
20. Feray S, Lubach J, Joshi GP, et al. PROSPECT guidelines for video-assisted thoracoscopic surgery: a systematic review and procedure-specific postoperative pain management recommendations. *Anaesthesia*. 2022;77(3):311-325. https://doi.org/10.1111/anae.15609
21. Arshad SA, Garcia EI, Bell C, et al. Multicenter assessment of cryoanalgesia use in minimally invasive repair of pectus excavatum: a 20-center retrospective cohort study. *Ann Surg*. 2023;277(6):e1373-e1379. https://doi.org/10.1097/SLA.0000000000005440
22. Lai K, Notrica DM, McMahon LE, et al. Cryoablation in 350 Nuss procedures: evolution of hospital length of stay and opioid use. *J Pediatr Surg*. 2023;58(8):1435-1439. https://doi.org/10.1016/j.jpedsurg.2022.10.051
23. Guitron J. Histopathologic study of the effects of surgically applied cryoanalgesia on intercostal nerves in a live porcine model. *J Clin Med*. 2024;13(11):3304. https://doi.org/10.3390/jcm13113304
24. Gwak MS, Yang M, Hahm TS, Cho HS, Cho CH, Song JG. Effect of cryoanalgesia combined with intravenous continuous analgesia in thoracotomy patients. *J Korean Med Sci*. 2004;19(1):74-78. https://doi.org/10.3346/jkms.2004.19.1.74

Key Words: cryoanalgesia, minimally invasive thoracic surgery, opioids, pain